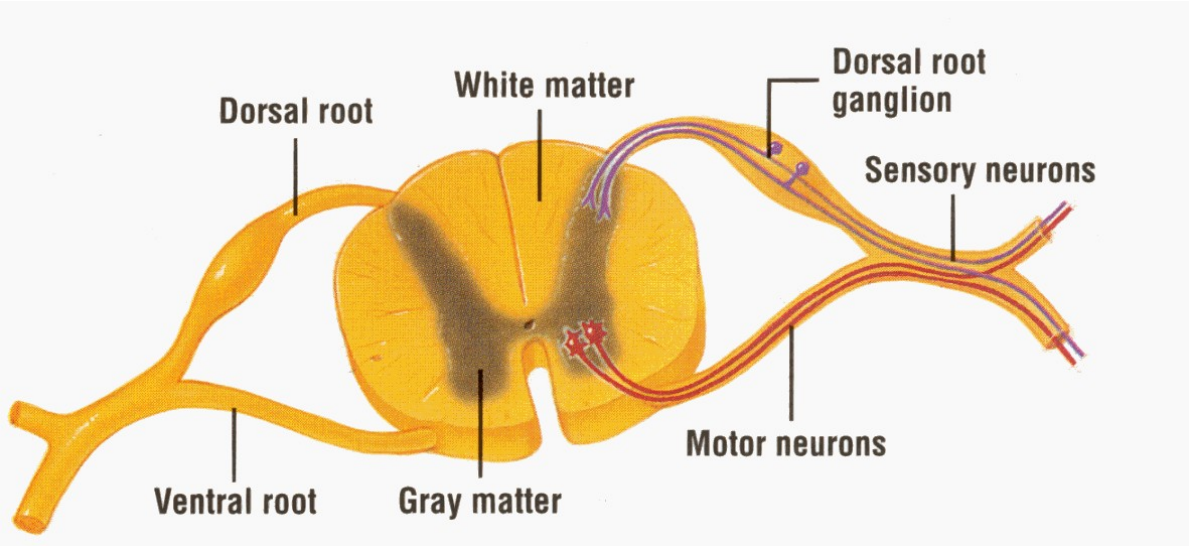
Nervous System Recap

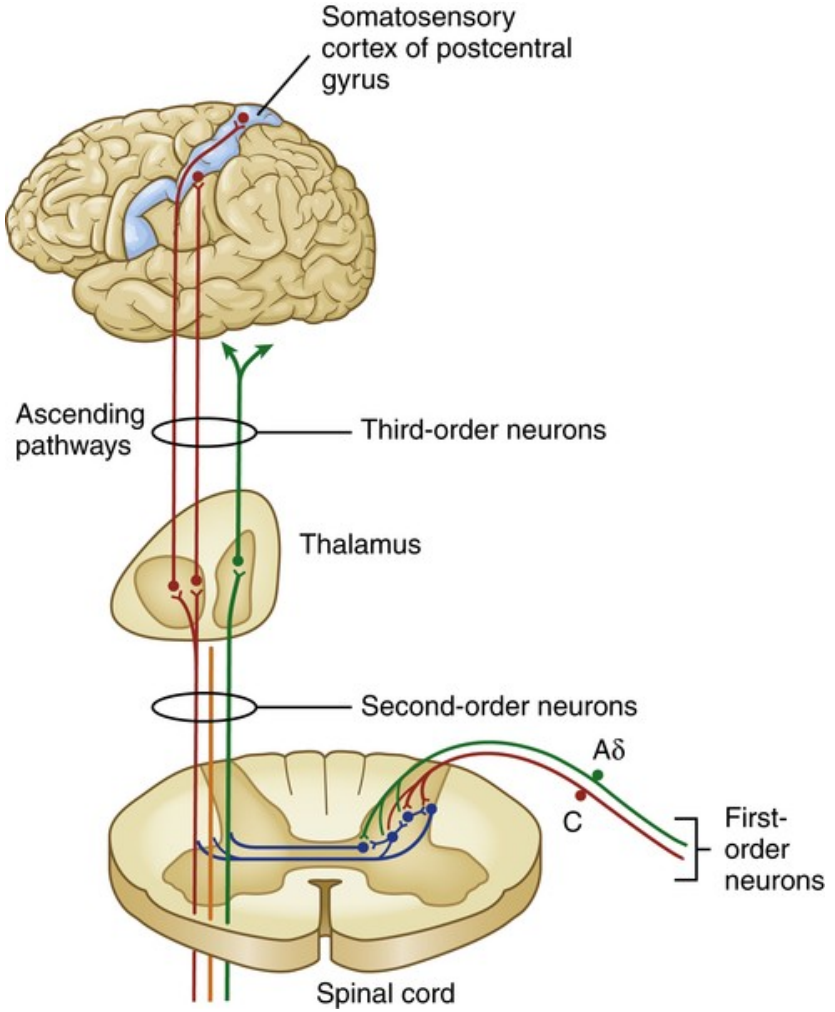
Divisions - Sensory

- Sensory:
  - detects information about internal and external environment.
  - Receptors detect information:
    - Rods and cones in retina
    - Ciliated cells in cochlea in ear
    - Taste buds on tongue
    - Olfactory buds in nasal cavity
    - Proprioceptive fibres in joints and muscles (position in space)
    - Receptors in skin for pressure, temperature and pain
    - Stretch receptors in muscles and walls of internal organs
    - Chemoreceptors/baroreceptors/thermoreceptors in hypothalamus and blood vessels

Divisions - Sensory

- Once information has been received about surrounding environment, signals are sent via sensory neurons to the CNS (brain and spinal cord)

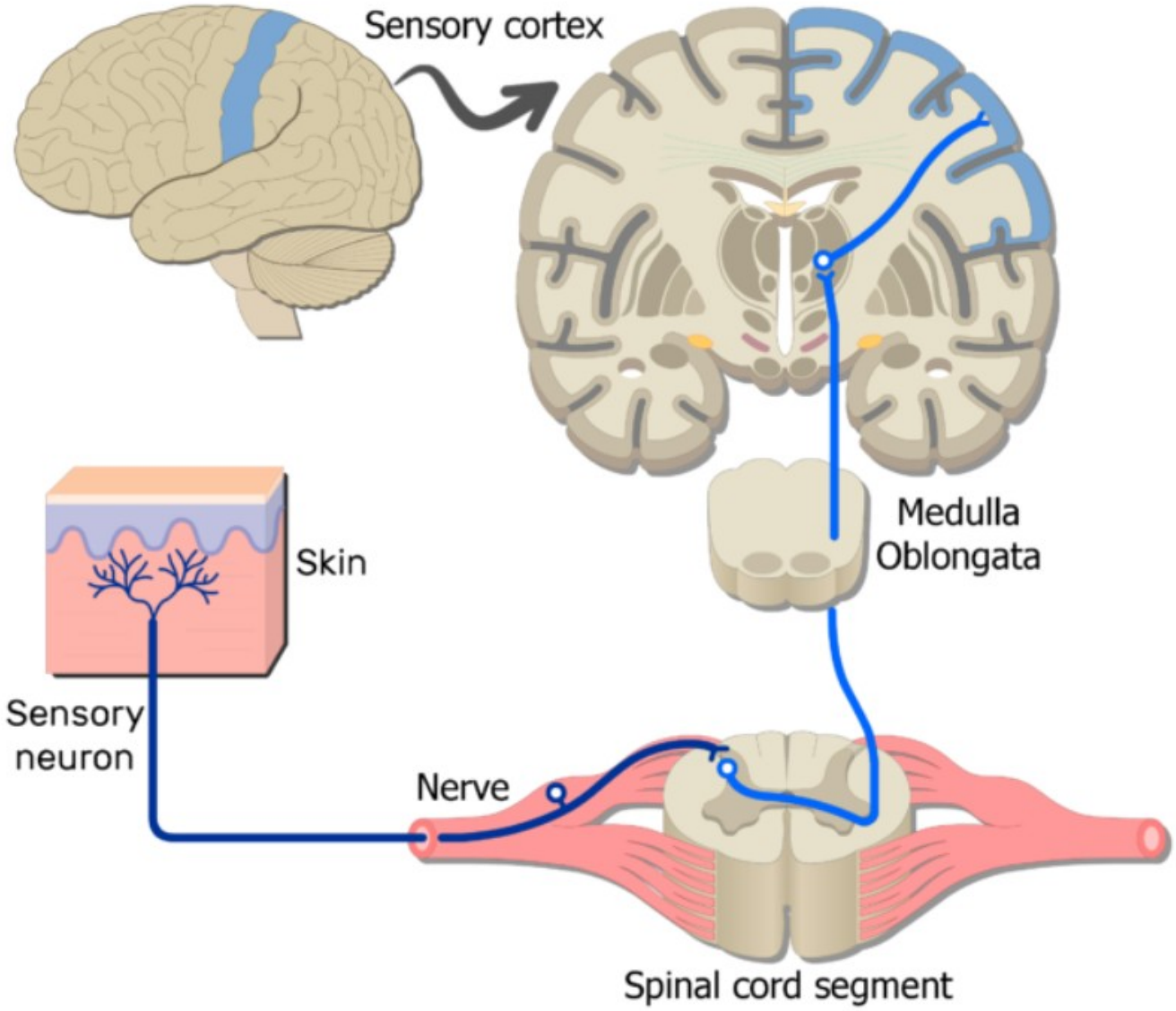




Sensory neurons that take info to spinal cord are:

- Pseudounipolar
- Have cell bodies in the dorsal root ganglion of the spinal cord
- ACh is the neurotransmitter at synapses

Divisions - Sensory

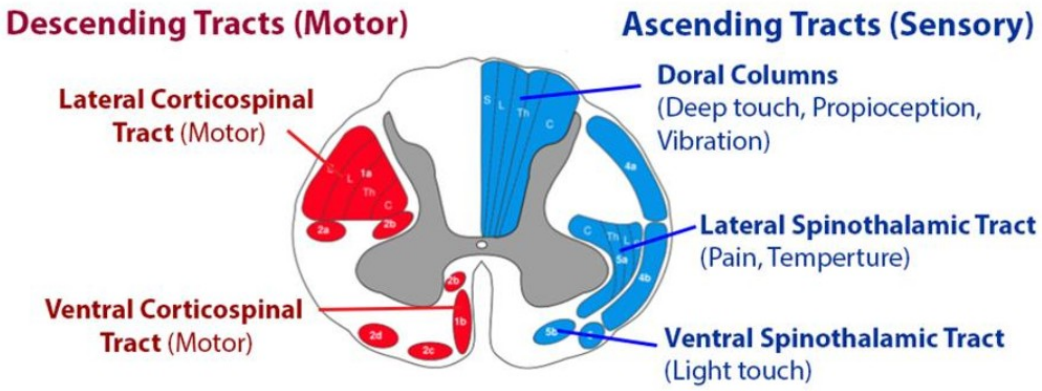


Divisions – CNS (Central Nervous System)

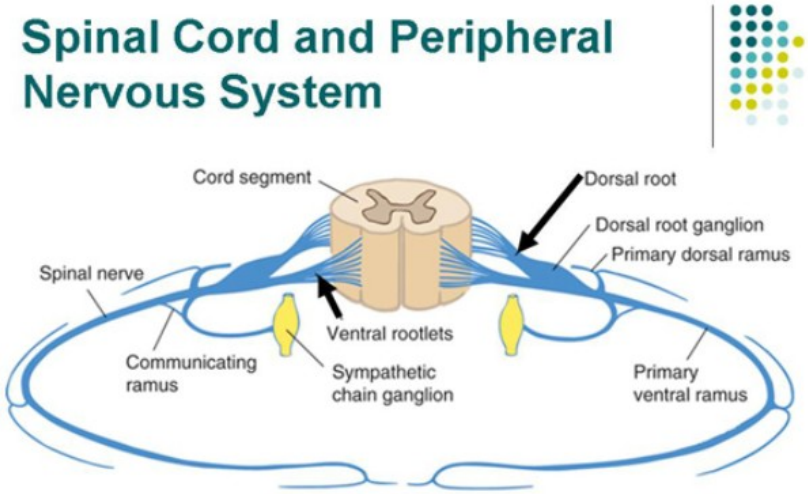
- Brain and spinal cord together make up the CNS
- Sensory information is received and processed, and then signals are sent out via the motor divisions for action.

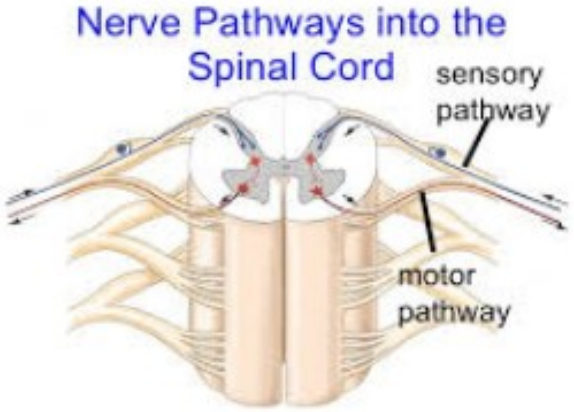
Divisions – CNS – Spinal Cord

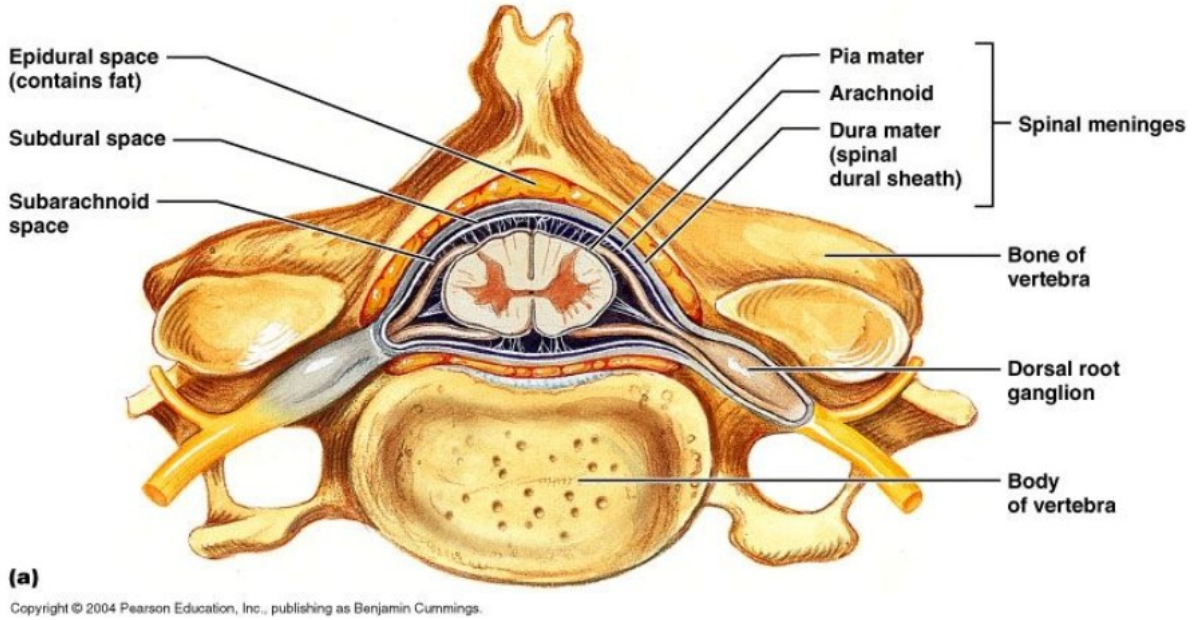
- Spinal Cord
  - Bundles of neurons (called tracts)
    - Sensory “ascending” tracts: take info up to brain from sensory neurons
    - Motor “descending” tracts: take info away from brain and back to effector muscles and glands
  - Many pathways involve a chain of neurons in spinal cord with multiple synapses
    - Eg sensory pathways from skin, up spinal cord to brain.



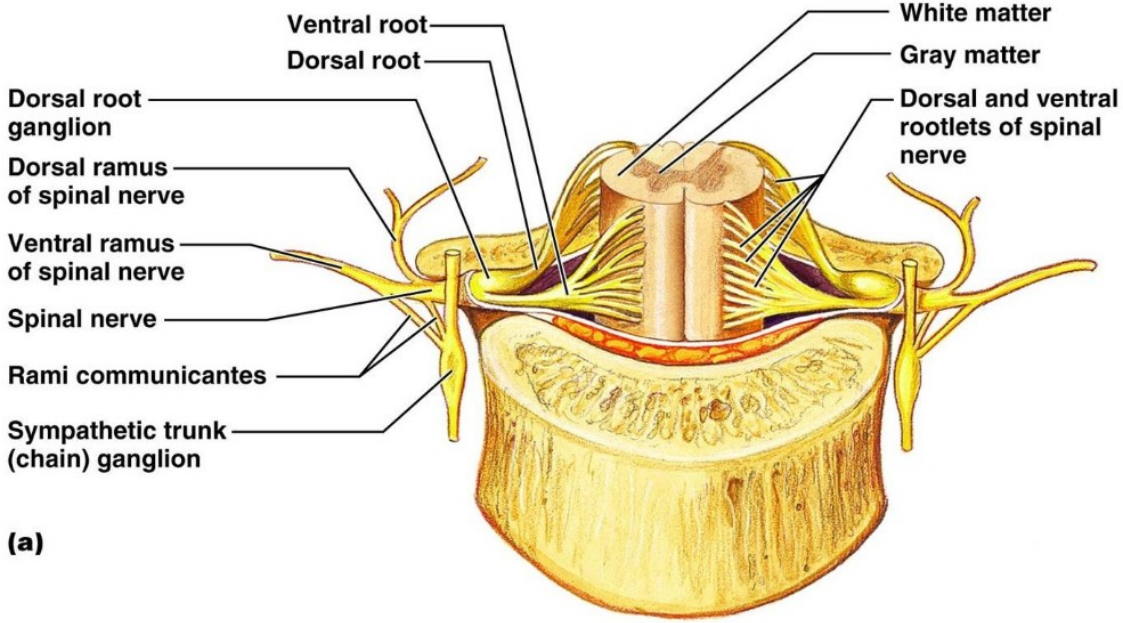
Divisions – CNS – Spinal Cc



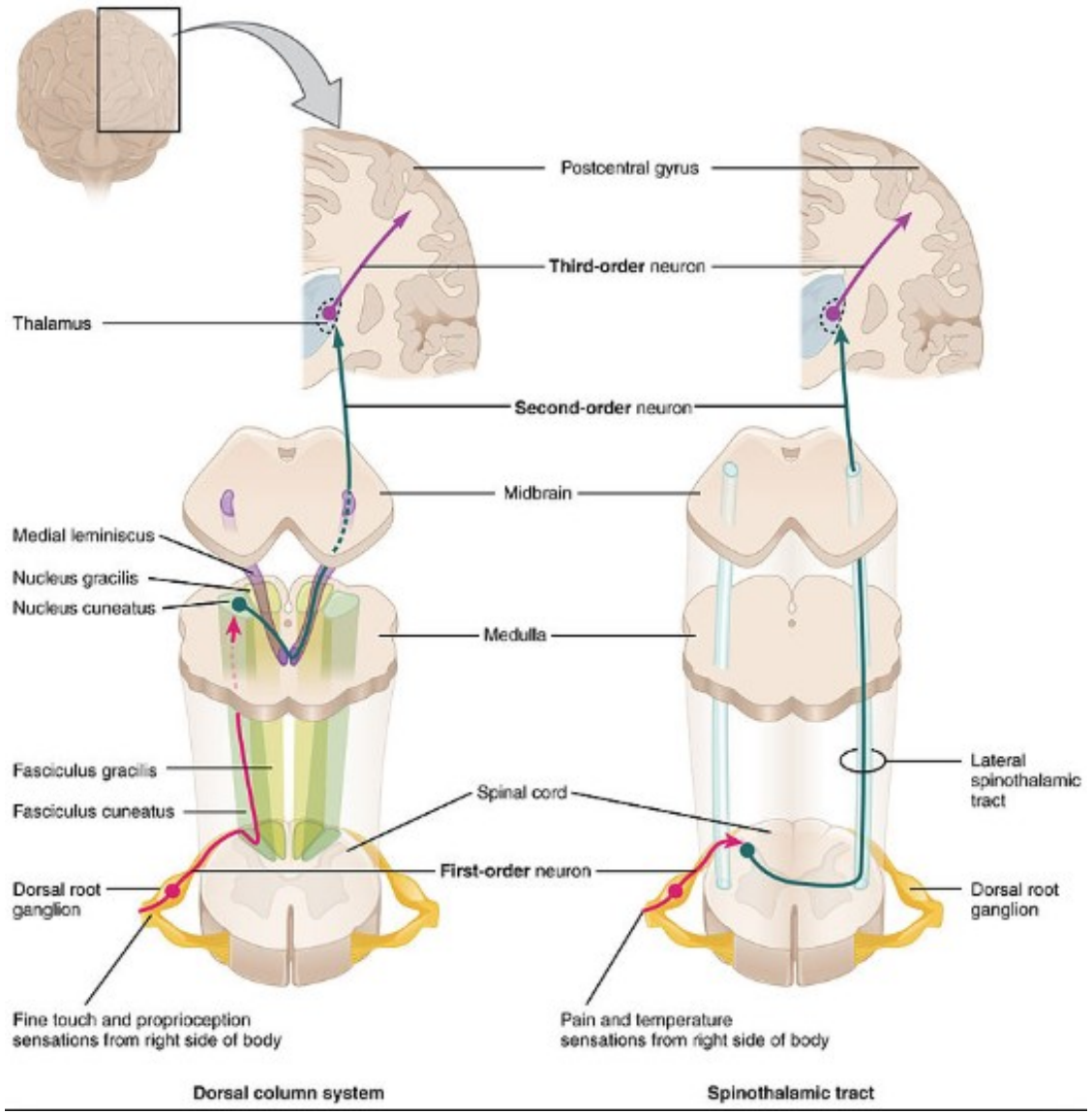




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Divisions – CNS – Pathways to and from Brain



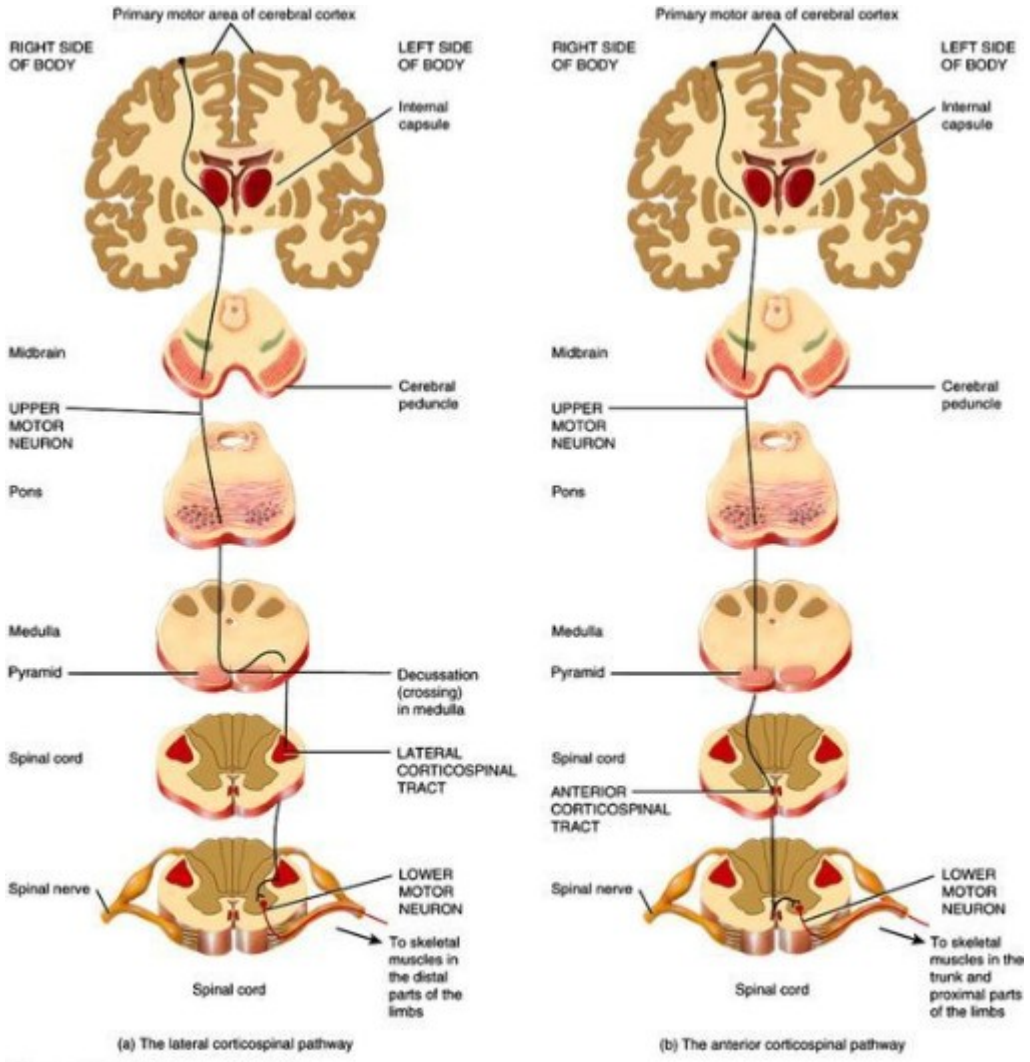
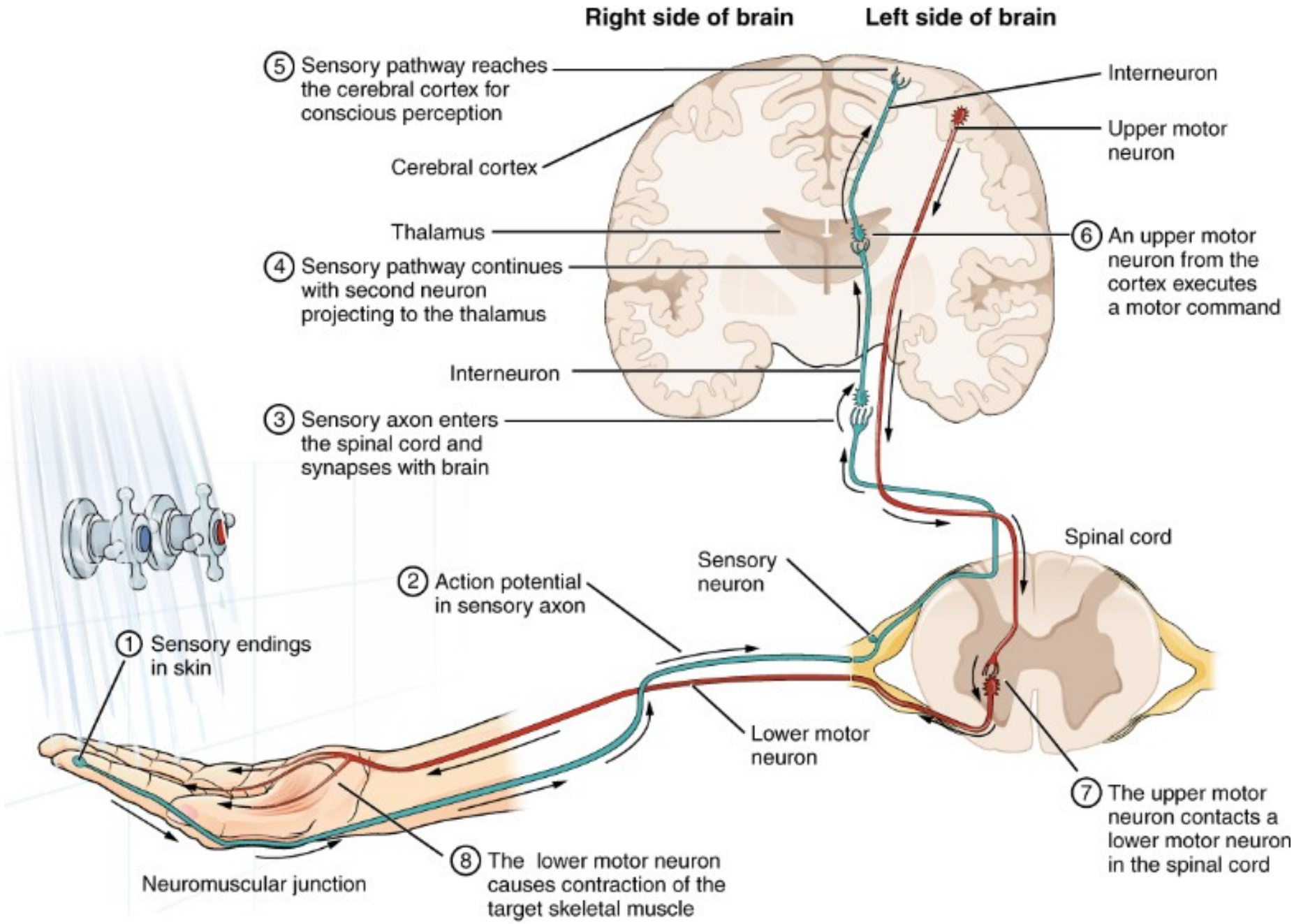
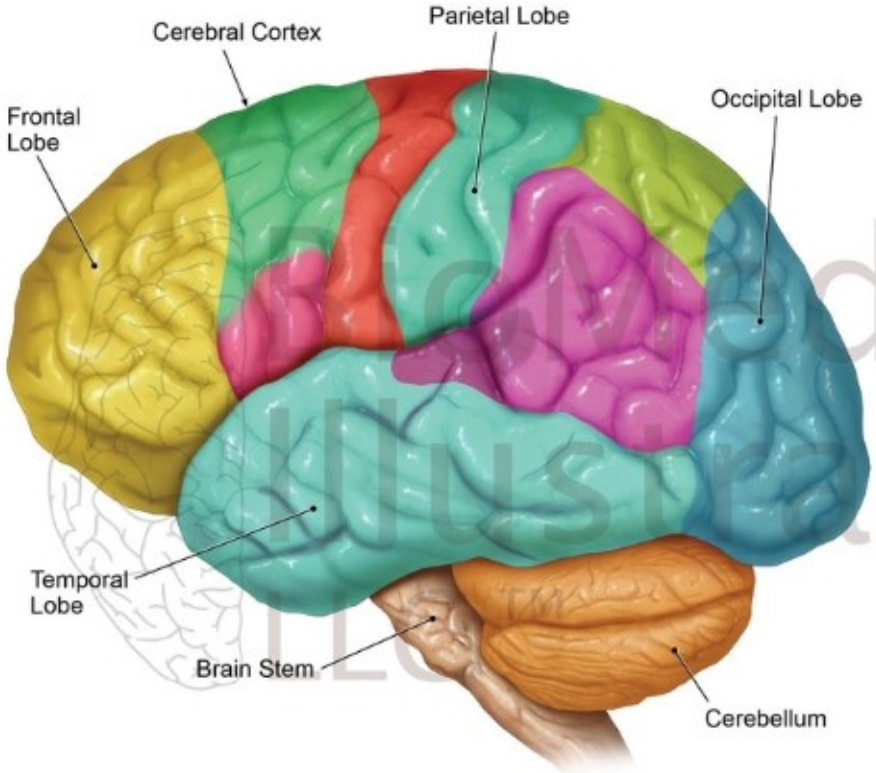


Figure 16.10 Tortora - PAP 12/e

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Right side of brain
Left side of brain
5 Sensory pathway reaches the cerebral cortex for conscious perception
Cerebral cortex
Thalamus
4 Sensory pathway continues with second neuron projecting to the thalamus
Interneuron
3 Sensory axon enters the spinal cord and synapses with brain
Interneuron
Upper motor neuron
6 An upper motor neuron from the cortex executes a motor command
Spinal cord
Sensory neuron
2 Action potential in sensory axon
1 Sensory endings in skin
Lower motor neuron
7 The upper motor neuron contacts a lower motor neuron in the spinal cord
Neuromuscular junction
8 The lower motor neuron causes contraction of the target skeletal muscle

Brain Surface Anatomy & Functions



Cerebral Functions

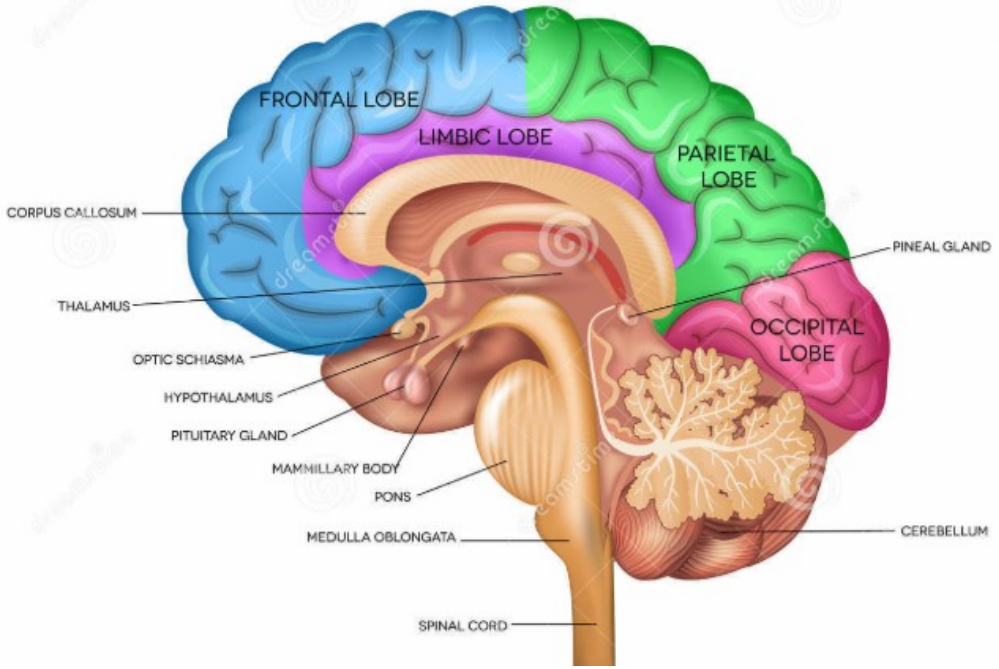
- **Higher Mental Function:** Problem Solving, Thinking, Planning, Judgement, Emotional Expression, Creativity, Behavioral Control
- **Motor Functions:** Orientation, Head and Eye Movements, Posture
- **Broca's Area:** Control of Muscles for Speech Production & Ability to Comprehend Grammatical Structure
- **Motor Functions:** Initiation of Voluntary Muscles, Movement
- **Sensory Functions:** Sensation from Skin and Muscles
- **Wernicke's Area:** Comprehension of Speech Sounds & Language
- **Auditory Functions:** Perception of Sounds
- **Tactile Perceptions:** Processing of Multi-Sensory Information
- **Visual Functions:** Coordination of Eye Movements, Perception, Image Recognition, Association, Visual Memory
- **Association Area:** Short Term Memory, Equilibrium, Emotion

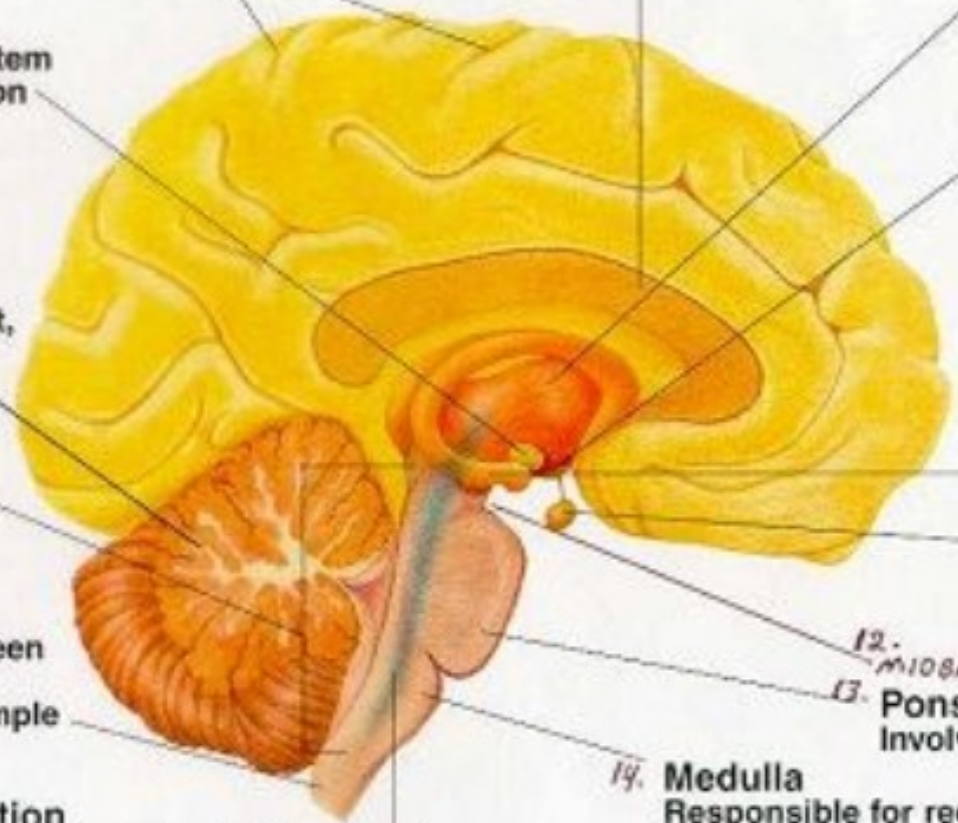
Cerebellar Functions

- **Motor Functions:** Coordinates Voluntary Movements: Posture, Balance, Coordination, & Speech

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ANATOMY OF THE BRAIN





1. **Cerebrum** CORTEX
Responsible for sensing, thinking, learning, emotion, consciousness, and voluntary movement

2. **Corpus callosum**
Bridge of fibers passing information between the two cerebral hemispheres

3. **Thalamus**
Relay center for cortex; handles incoming and outgoing signals

4. **Hypothalamus**
Responsible for regulating basic biological needs: hunger, thirst, temperature control

5. **Pituitary gland**
"Master" gland that regulates other endocrine glands

6. **Hippocampus**
Part of limbic system involved in learning and memory

7. **Amygdala**
Part of limbic system involved in emotion and aggression

8. **Cerebellum**
Structure that coordinates fine muscle movement, balance

9. **Brainstem**

10. **Spinal cord**
Responsible for transmitting information between brain and rest of body; handles simple reflexes

11. **Reticular formation**
Group of fibers that carry stimulation related to sleep and arousal through brainstem

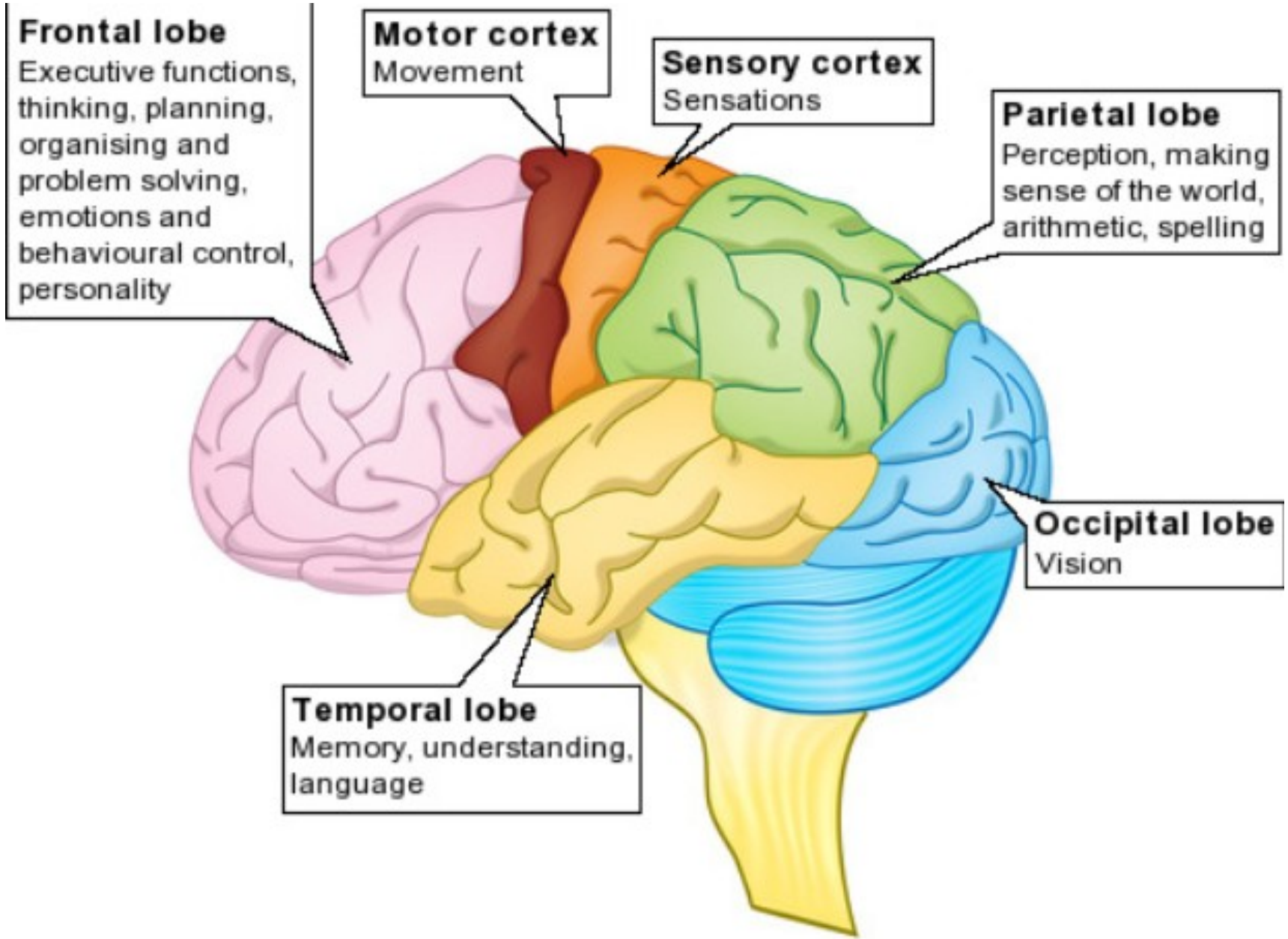
12. MIDBRAIN

13. **Pons**
Involved in sleep and arousal

14. **Medulla**
Responsible for regulating largely unconscious functions such as breathing and circulation

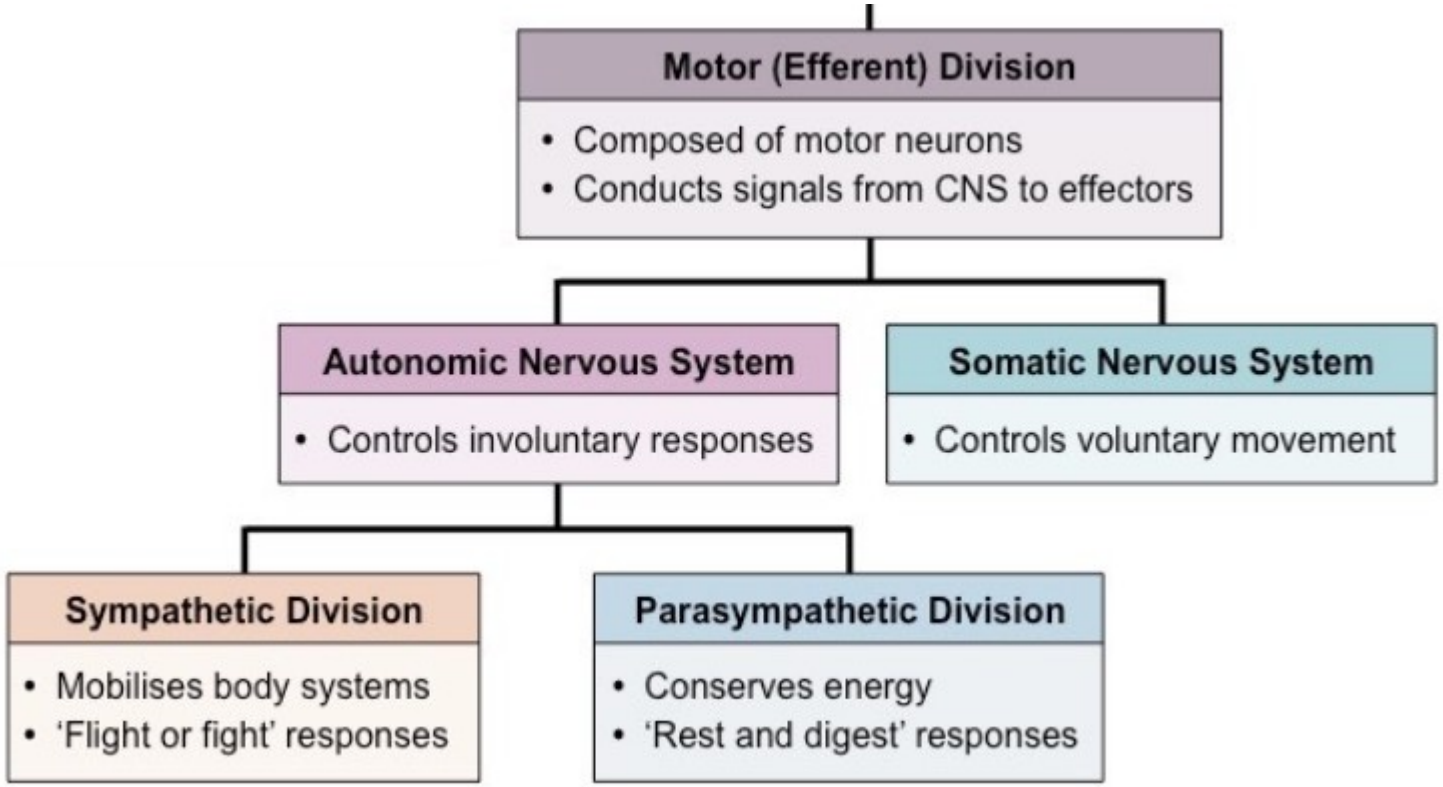
Structures and areas in the human brain

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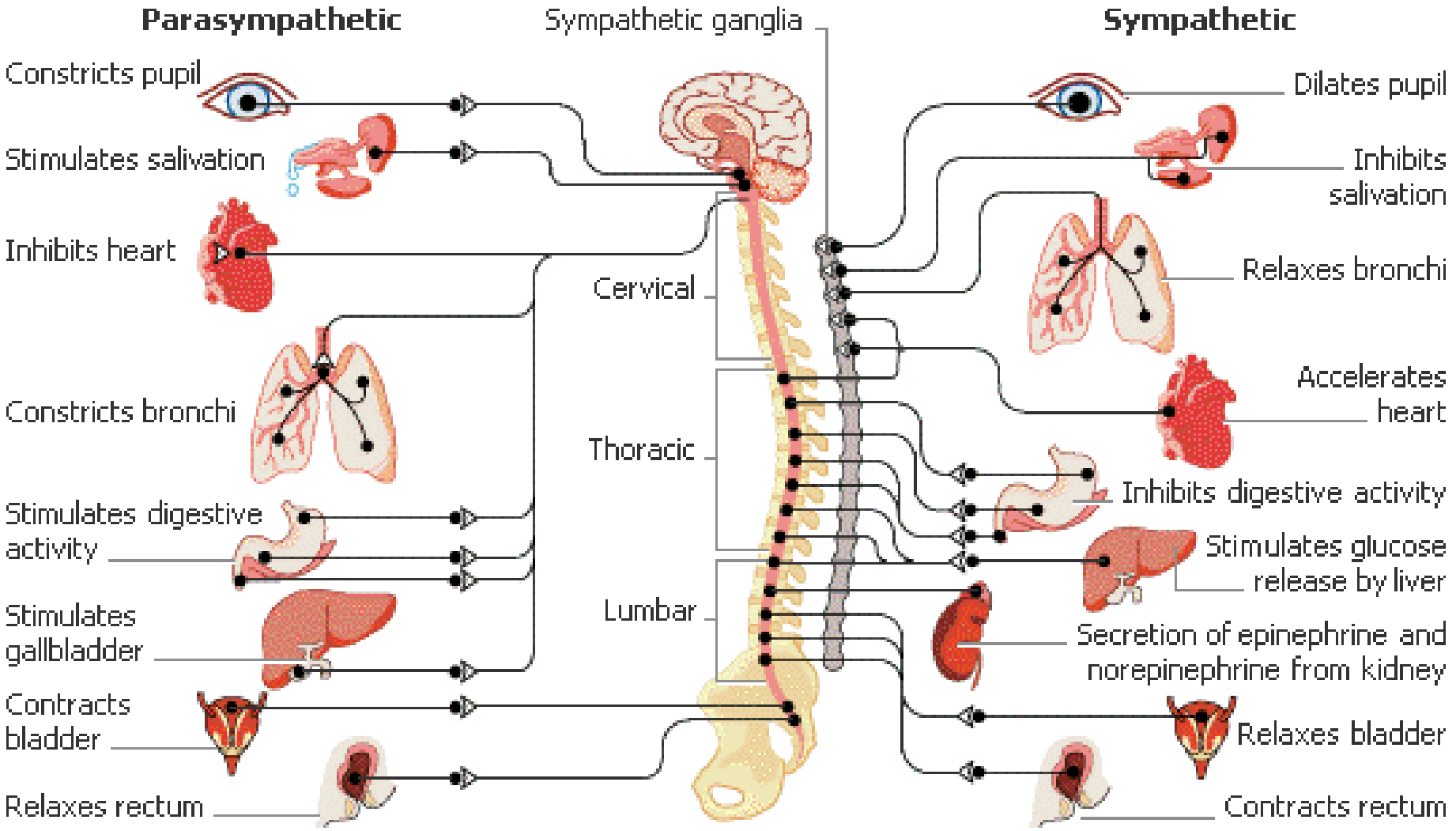

Frontal lobe
Executive functions, thinking, planning, organising and problem solving, emotions and behavioural control, personality
Motor cortex
Movement
Sensory cortex
Sensations
Parietal lobe
Perception, making sense of the world, arithmetic, spelling
Occipital lobe
Vision
Temporal lobe
Memory, understanding, language

Divisions – Motor

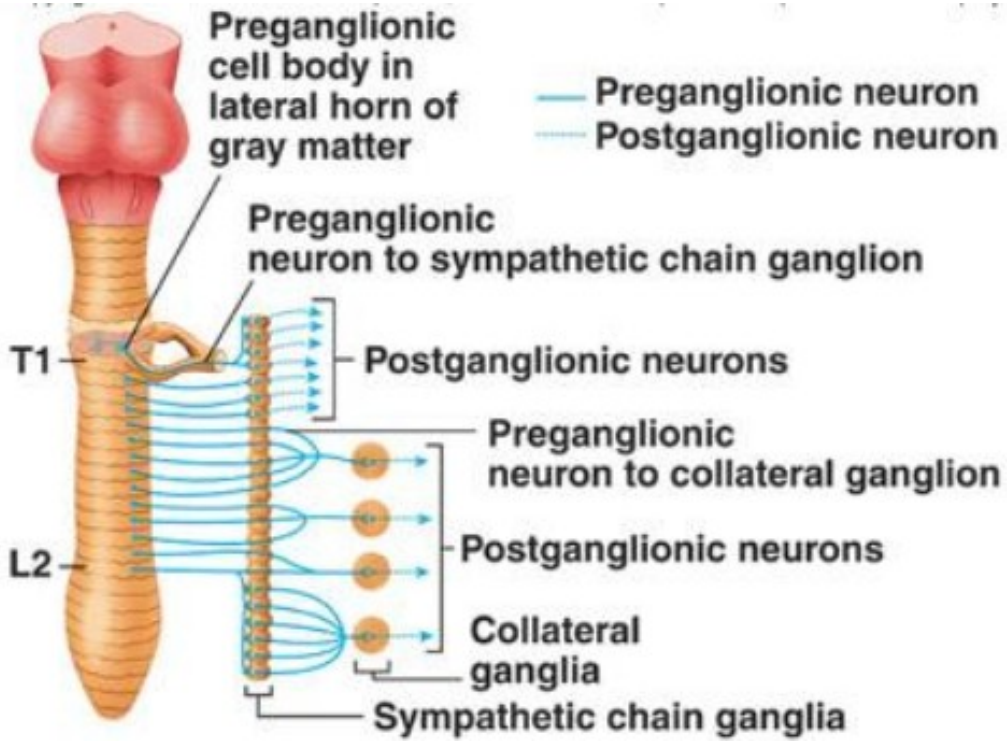
- Output/Action
- Messages from CNS (brain and spinal cord) sent to muscles and glands
- Divisions within the Motor NS

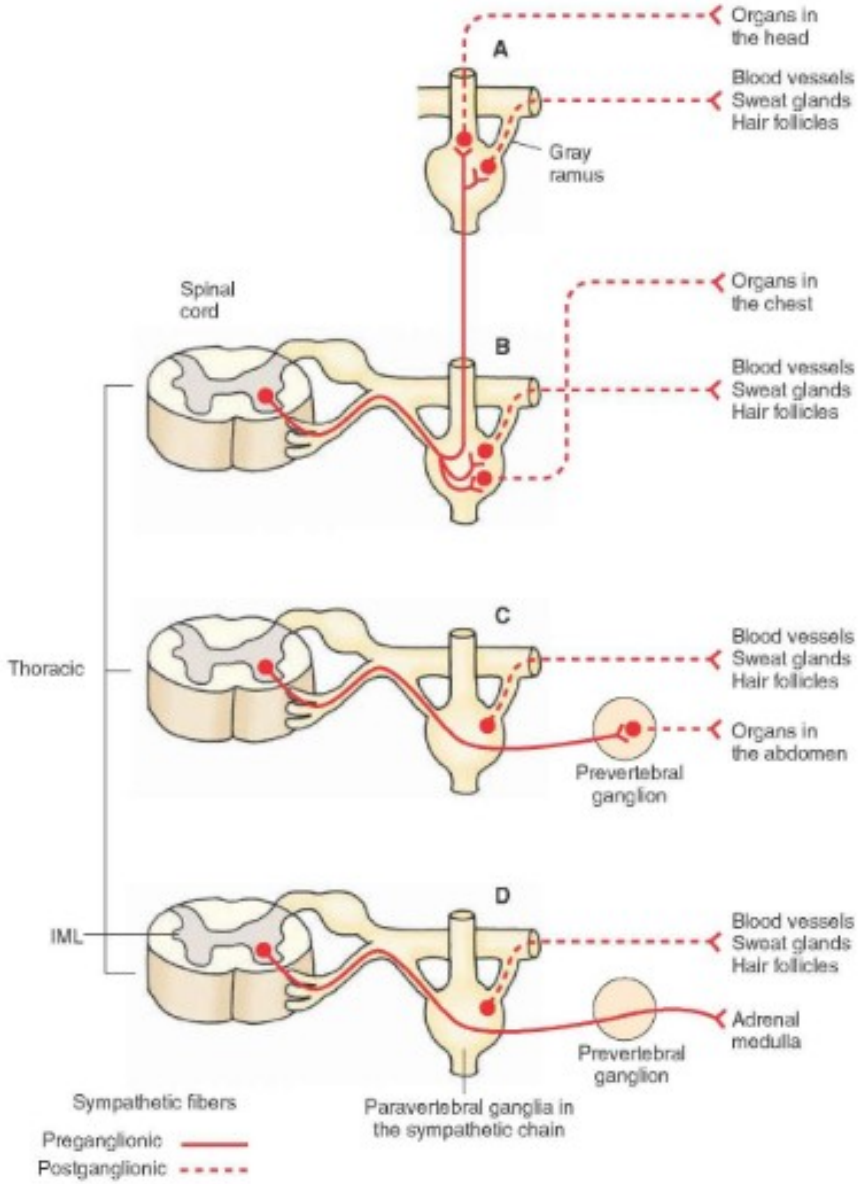


Divisions- Motor > Autonomic

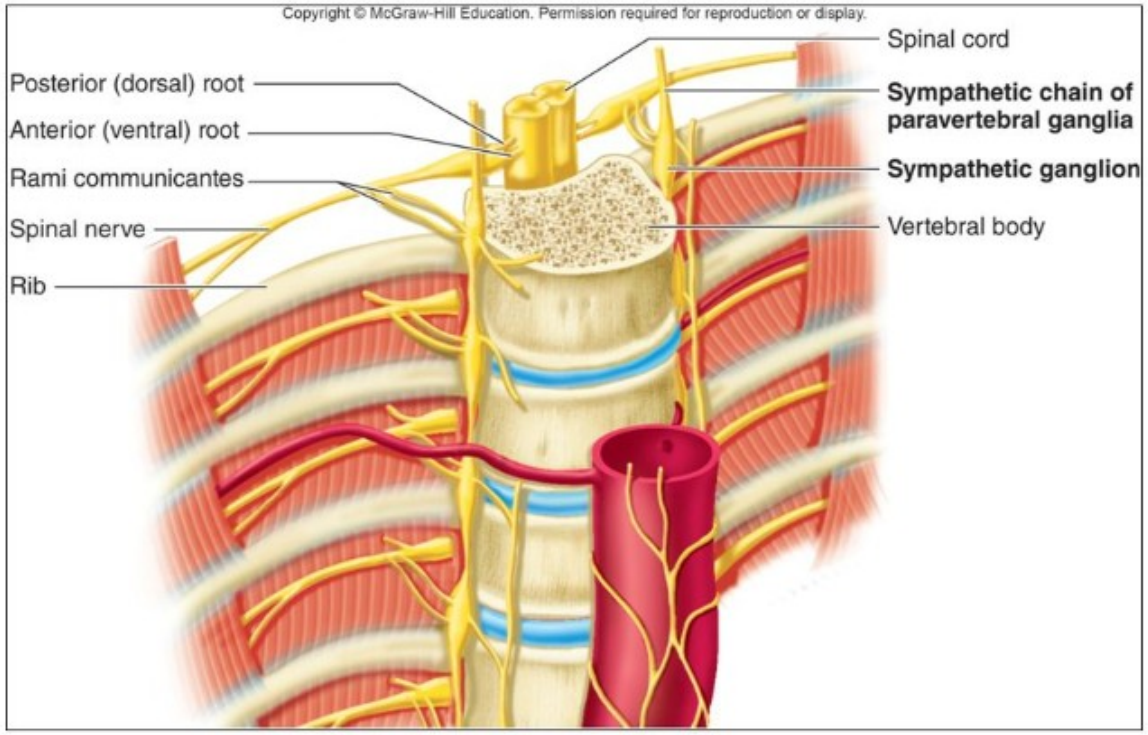


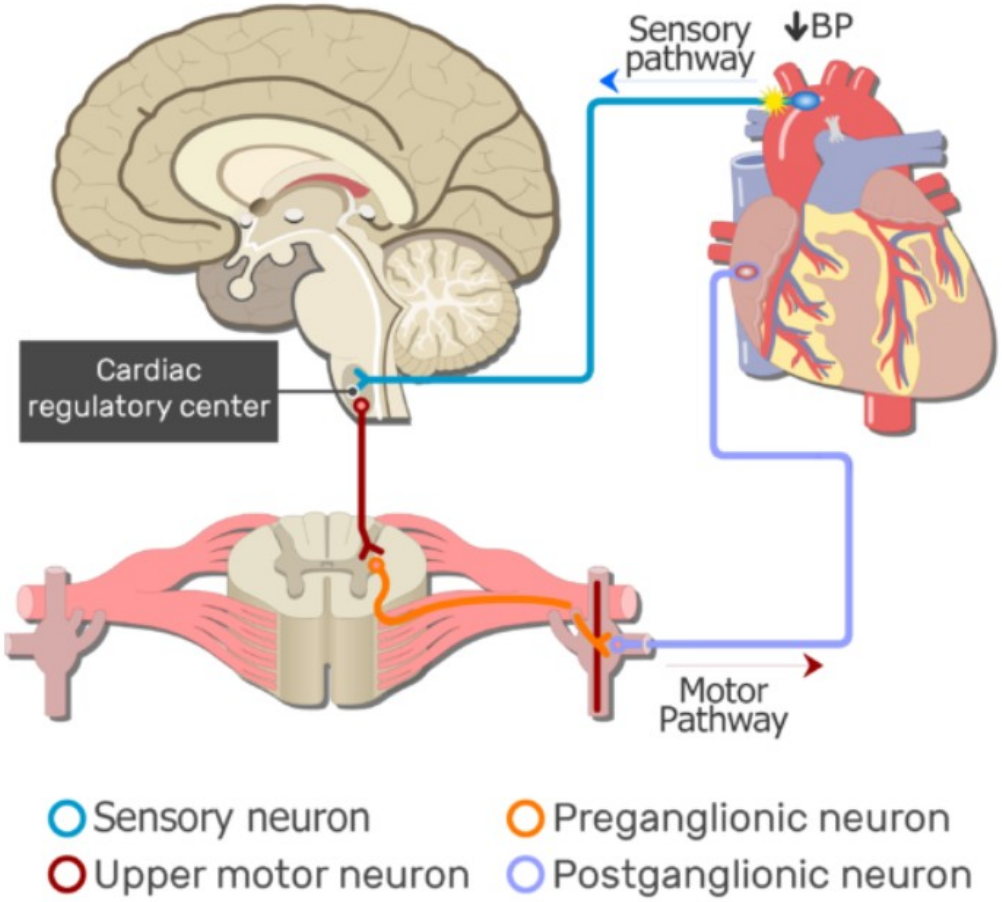
Divisions – Motor > Autonomic >Sympathetic



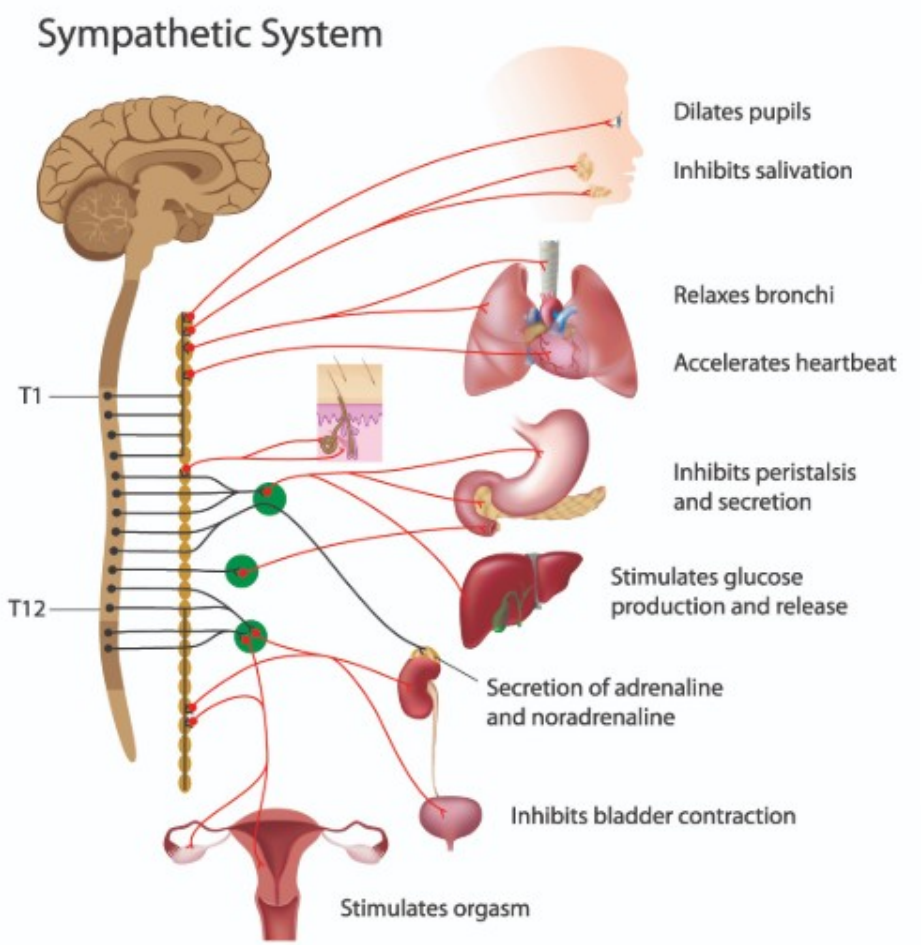


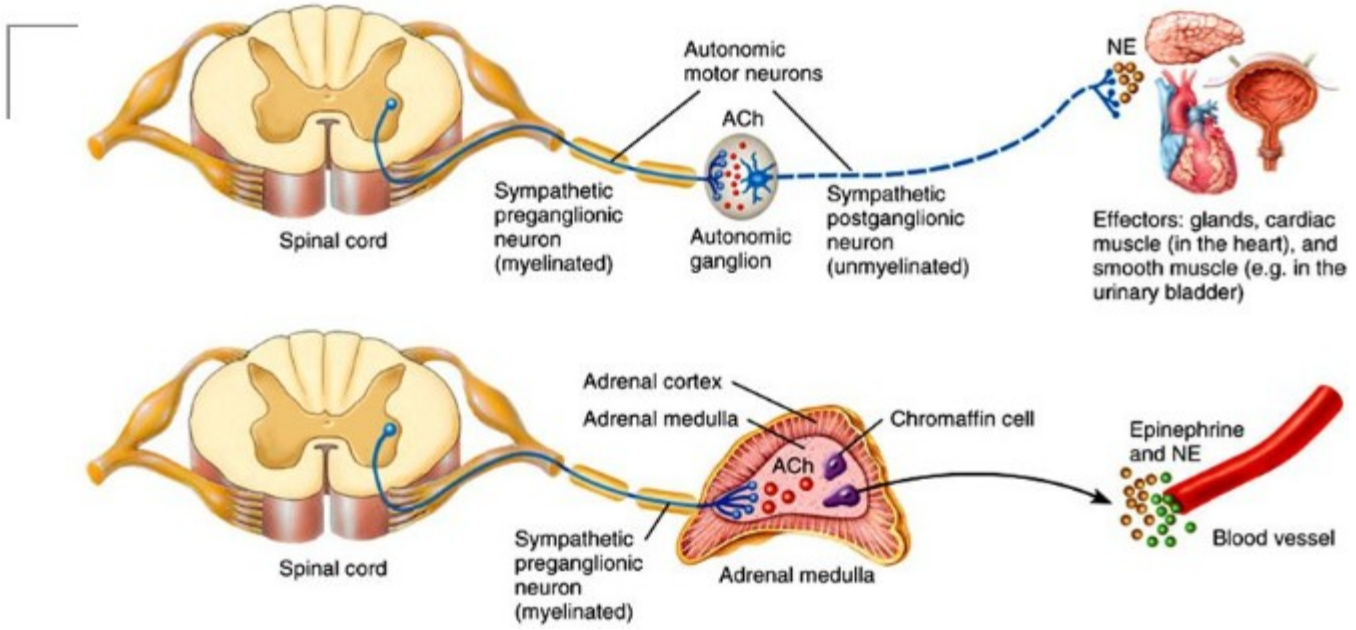
Divisions – Motor > Autonomic >Sympathetic



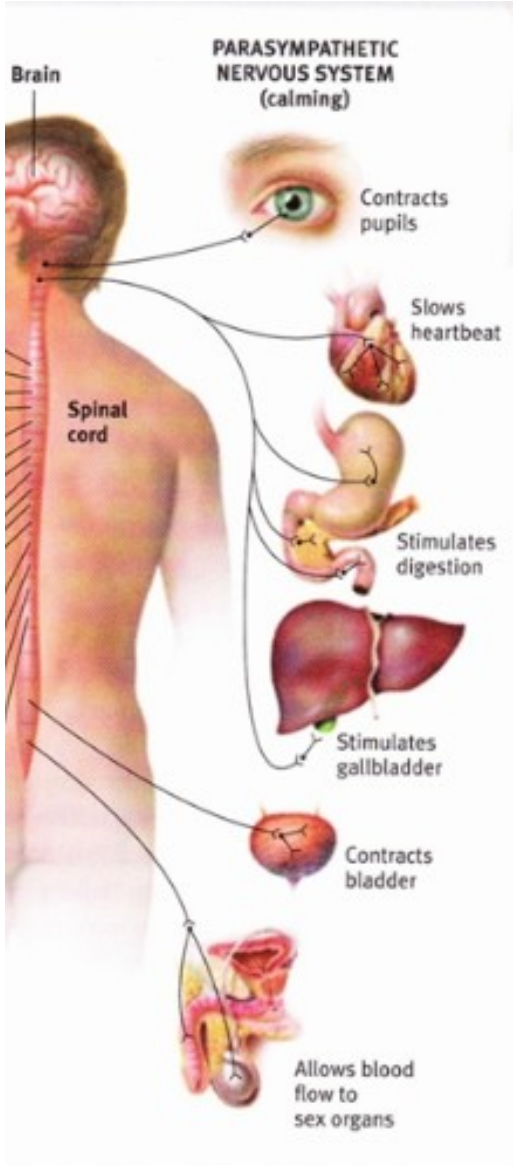


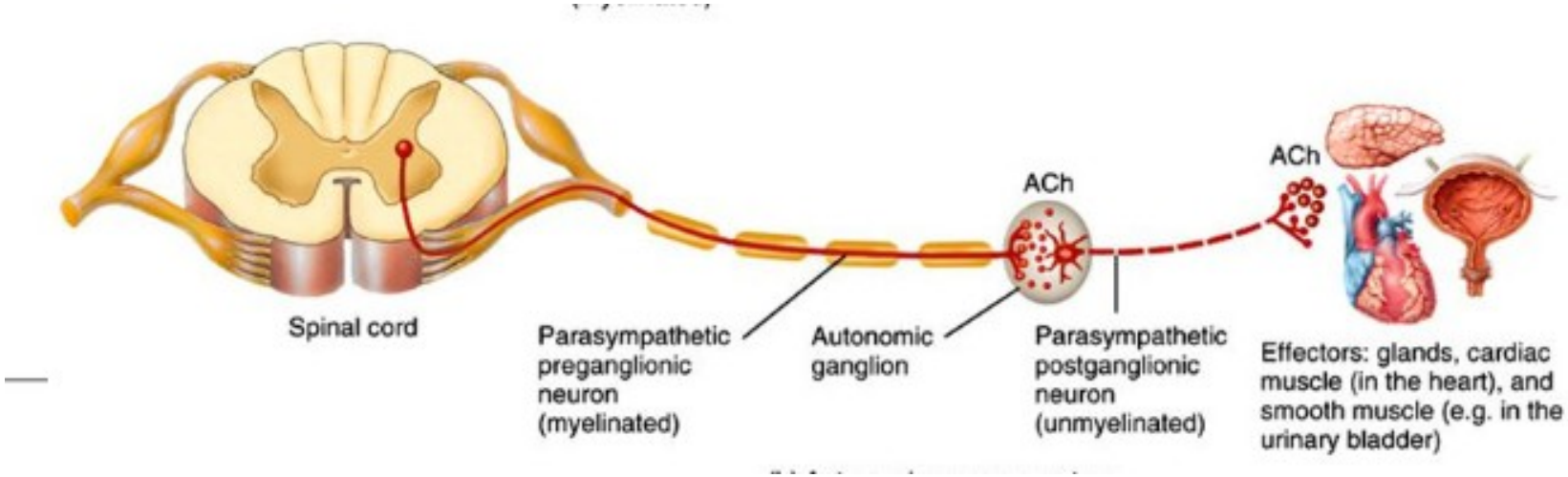
Divisions – Motor>Autonomic>Sympathetic





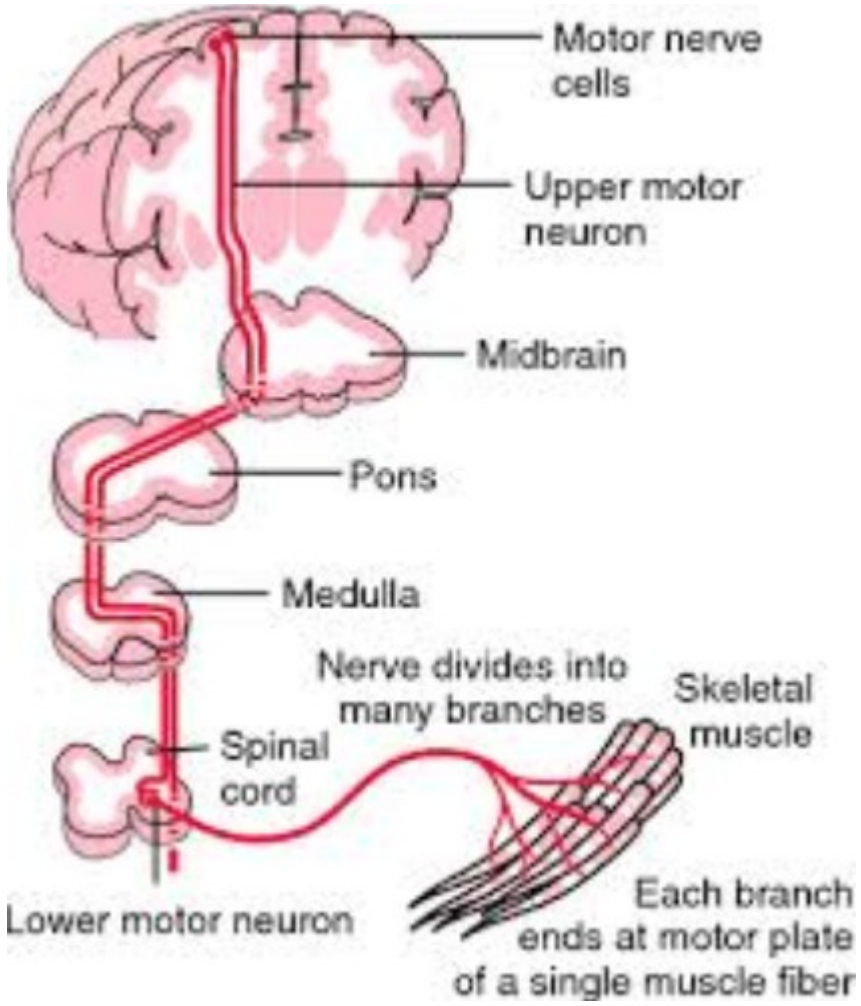
Divisions- Motor>Autonomic>Parasympathetic

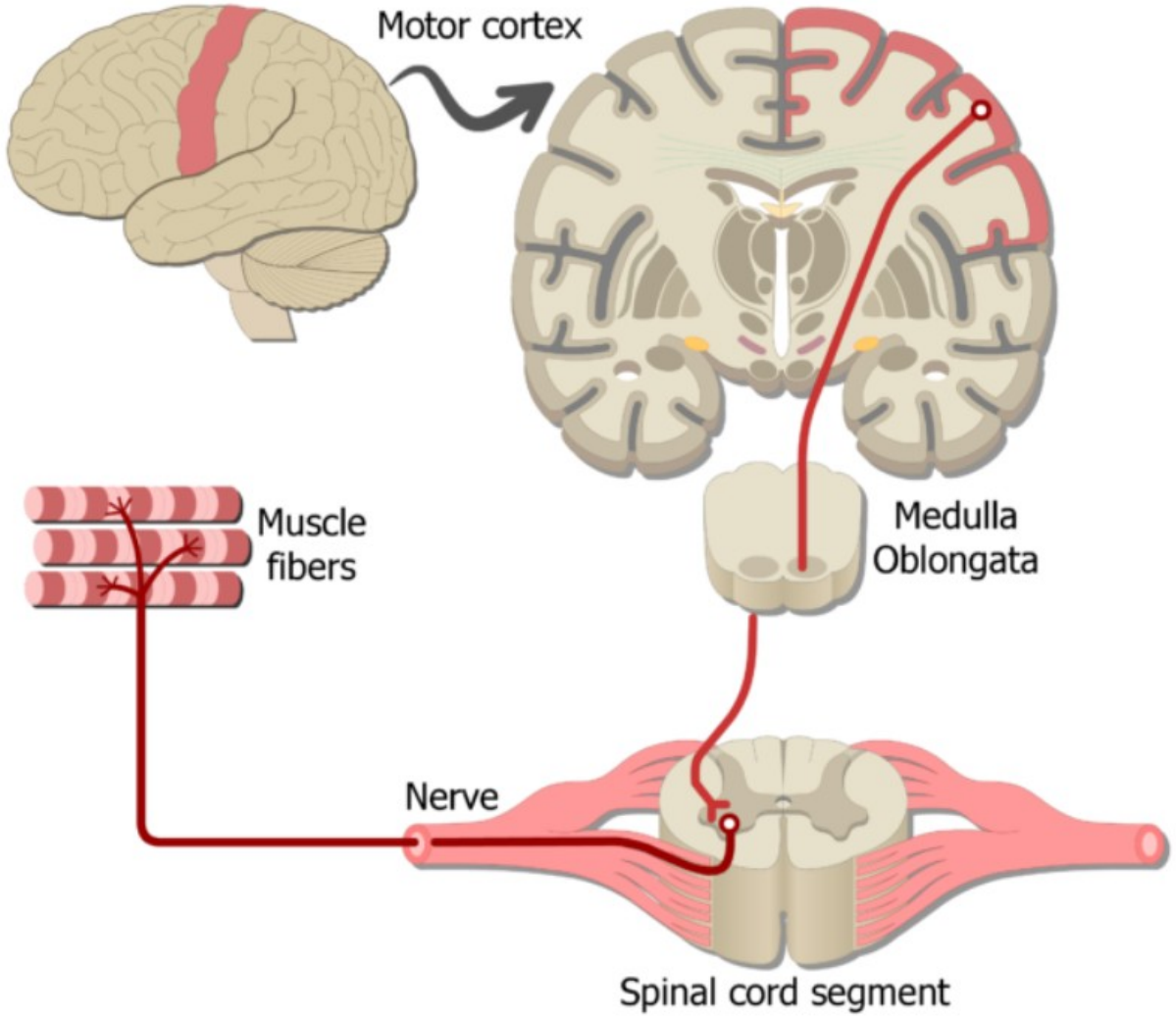




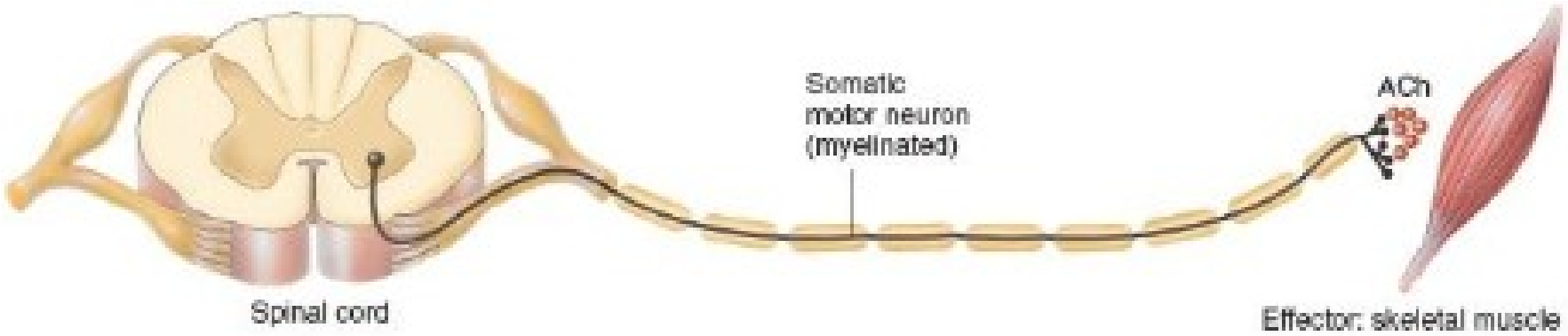
Note: Parasympathetic neurons exit spinal cord at Brain, Cervical Spine and Sacral (pelvic) spine only

Divisions – Motor >Somatic





Divisions – Motor>Somatic



(a) Somatic nervous system

Motor Pathways of the SNS and ANS

